

Executive Summary

Surrogate Model Validation Report

UCFATIGUE_1

July 17, 2026

REQUIREMENTS PARTIALLY MET

Best model: GradientBoosting Avg R²: 0.9643 Q90 pass: 5/7 KS pass: 2/7

1. Project Overview

Use case	UCFATIGUE_1
Models	MLP, GradientBoosting
Outputs	7 (1g, Vertical maneuver, Vertical gust, Turn, Frontal gust, n0, Giro)
Inputs	FLAP, Altitude, TAS, Mass, q, gamma, Type_segment, Xcg(%CMA)
Train / Val / Test	— / — / — rows
Requirement	Q90 relative error < 10% per output

2. Model Comparison (Test Set)

Model	Avg R ²	Q90 pass (7 outputs)	KS pass (7 outputs)
MLP	0.5237	5/7	7/7
GradientBoosting	0.9643	5/7	2/7

Q90 target: < 10% relative error at 90th percentile. KS test: $p \geq 0.05$ means train and test residuals follow the same distribution (no overfitting).

3. Metrics per Output (Q90 target < 10%)

Output	MLP		GradientBoosting	
	R ²	Q90	R ²	Q90
1g	0.9160	0.047	0.9797	0.026
Vertical maneuver	0.9116	0.055	0.9723	0.032
Vertical gust	0.9377	0.092	0.9914	0.026
Turn	0.7544	0.078	0.9286	0.035
Frontal gust	0.9794	11.330	0.9964	0.280
n0	-0.7292	0.072	0.9317	0.015
Giro	-0.1042	0.901	0.9499	0.222

pass Q90 < 10%

fail Q90 ≥ 10%

R² ≥ 0.95 0.80 ≤ R² < 0.95 R² < 0.80

4. Overfitting Check (KS Test p-values)

H_0 : train and test residuals follow the same distribution. Rejection ($p < 0.05$) indicates overfitting.

Output	MLP	GradientBoosting
1g	0.473	0.004
Vertical maneuver	0.180	0.161
Vertical gust	0.635	0.042
Turn	0.078	0.044
Frontal gust	0.736	0.974
n0	0.378	0.032
Giro	0.781	0.017

5. Train/Test Split Quality

Residual voxel proportion	0.000 ×
Valid test proportion	0.931
Phacking test proportion	0.006
Isolated test proportion	0.063
Chi ² p-value	0.7969 ✓ (≥ 0.05)

6. Recommendation

- Recommended model: GradientBoosting (highest average $R^2 = 0.9643$, Q90 pass rate 5/7, KS pass rate 2/7).
- Split quality is excellent: no residual voxels, no phacking artefacts.
- Deploy `pipeline_UCFATIGUE_1_GradientBoosting.pkl` for inference.